Quasi-Black Holes and the Bost-Connes System

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Abstract

The Bost-Connes (BC) algebra is a C^*-dynamical system whose partition function is the Riemann ζ function; its unique phase transition at $\beta = 1$ is a product of the thermodynamic limit $N \to \infty$. Physical compact objects in nature have finitely many degrees of freedom $N < \infty$, so their truncated partition function $Z_N(\beta)$ is analytic for all $\beta > 0$ — no pole, no sharp phase transition, no sharp horizon. We define a **quasi-black hole** as a finite system with $N \gg 1$ and effective inverse temperature β lying in the BC crossover region, and derive its formation, properties, evaporation, and endpoint from the BC algebra.

Main results: (i) the discrete entropy of a quasi-black hole is $S_{\mathrm{BC}} = \psi(N) = \log \mathrm{lcm}(1, \ldots, N)$, where ψ is the Chebyshev function [28], with an exact expansion containing ζ-zero corrections; (ii) the direction and rate of evaporation are determined by KMS detailed balance; (iii) evaporation terminates at the stable Planck ground state $N = 1$. The observable universe itself satisfies the quasi-black hole definition ($N_{\mathrm{univ}} \approx 10^{122}$), and stellar-mass objects are its subsystems. The Bekenstein-Hawking entropy $4\pi N$ and the Hawking temperature are recovered as approximations in the $N \to \infty$ continuum limit.

Contents

1 Introduction

1.1 Two Fundamental Difficulties with GR Black Holes

General relativity's predictions for black holes represent one of the deepest achievements of twentieth-century physics. At the theoretical level, however, the GR picture of black holes faces two fundamental difficulties.

Difficulty I: Singularities The GR singularity theorems (Penrose, 1965 [6]) predict that, under generic initial conditions, gravitational collapse inevitably produces singularities—points in spacetime where curvature diverges. General relativity itself breaks down at singularities, requiring recourse to quantum gravity. Recent LIGO/VIRGO/KAGRA detections of binary black hole merger gravitational waves [18] and EHT shadow images of M87* and Sgr A* [19, 20] are in excellent agreement with GR predictions. Yet at the theoretical level, fifty years of quantum gravity proposals have each provided their own interpretation without reaching a unified consensus.

Difficulty II: The Information Paradox Hawking (1975) [9] showed that black hole radiation is exactly thermal, causing a pure state to evolve into a mixed state and violating the unitarity of quantum mechanics. Page (1993) [12] noted that, if unitarity holds, the radiation entropy should first increase and then decrease (the Page curve). Recent developments—the complementarity principle [13], the firewall argument [15], and the island formula [16, 17]—have brought new progress, but a complete resolution remains elusive.

The common root of both difficulties: GR treats black holes as products of the $N \to \infty$ thermodynamic limit—infinitely many degrees of freedom, a sharp horizon, and a sharp phase transition. Singularities and the information paradox are both mathematical consequences of this limit.

1.2 The BC Framework

The Bost-Connes (BC) algebra [1, 2] is a C^*-dynamical system whose partition function is the Riemann ζ function and whose KMS states are parametrized by the arithmetic structure of the integers [3]. The key observation underlying the BC framework is that no system in nature has $N = \infty$. The truncated partition function for finite N, $Z_N(\beta) = \sum_{n=1}^{N} n^{-\beta}$, is analytic for all $\beta > 0$—no pole, no sharp phase transition, no sharp horizon. Singularities are absent ($n \geq 1$ ensures finite curvature analogues), and the information paradox dissolves (no sharp horizon to trap information).

This paper derives the definition, formation, properties, evaporation, and endpoint of quasi-black holes entirely from BC first principles, without invoking any continuum GR concept (Tolman temperature, Schwarzschild radius, Hawking radiation, etc.). Continuum GR appears only in Sec. 9 as the $N \to \infty$ limiting approximation.

Honesty grading. This paper uses three levels of annotation: **[A]** rigorous theorem (proved in the original BC literature or in number theory); **[A*]** semi-rigorous derivation (reasonable assumptions with an explicit argument chain); **[B]** physically motivated (conjecture, awaiting rigorous justification).

2 Basic Quantities of the BC Algebra

2.1 The BC Hilbert Space and Hamiltonian [A]

Theorem 2.1 ([**A**], Bost-Connes 1995 [1])**.** *The quantum-mechanical ingredients of the Bost-Connes system are:*

- ***Hilbert space:*** *$\mathcal{H} = \ell^2(\mathbb{Z}_{>0})$, with orthonormal basis $\{|n\rangle\}_{n\geq 1}$ (indexed by the positive integers);*

- ***BC Hamiltonian:***
$$H_{\mathrm{BC}}\,|n\rangle = (\log n)\,|n\rangle, \qquad n = 1, 2, 3, \ldots$$
with spectrum $\{\log n : n \geq 1\} = \{0, \log 2, \log 3, \log 4, \ldots\}$;

- ***Time evolution:*** *$\sigma_t(a) = e^{iH_{\mathrm{BC}}t} a\, e^{-iH_{\mathrm{BC}}t}$, a one-parameter group of $*$-automorphisms of the BC algebra $\mathcal{A}_{\mathrm{BC}}$;*

- ***Partition function*** *($\beta > 1$):*
$$Z(\beta) = \mathrm{tr}\big(e^{-\beta H_{\mathrm{BC}}}\big) = \sum_{n=1}^{\infty} e^{-\beta \log n} = \sum_{n=1}^{\infty} \frac{1}{n^{\beta}} = \zeta(\beta).$$

Remark 2.2. H_{BC} is intrinsic to the BC algebra: its eigenvalues are determined by the algebraic structure of the integers, defined independently of any spacetime geometry. The energy $\log n$ of the state $|n\rangle$ attains its minimum value of zero at $n = 1$ (the ground state), with no singularity ($n \geq 1$, excluding $n = 0$).

2.2 The Truncation Parameter N [A]

Definition 2.3 (Truncation Parameter, **[A]**)**.** The **truncation parameter** N is defined as the index of the highest currently excited state in the BC system. The Hilbert space is truncated to
$$\mathcal{H}_N = \mathrm{span}\{|1\rangle, |2\rangle, \ldots, |N\rangle\},$$
and the **truncated partition function** is
$$Z_N(\beta) = \sum_{n=1}^{N} \frac{1}{n^{\beta}}.$$
N is a parameter of the BC system itself, not imported from GR—it describes how many discrete states the system has excited.

Remark 2.4. $N = \infty$ corresponds to the full infinite-dimensional $\ell^2(\mathbb{Z}_{>0})$, giving $Z(\beta) = \zeta(\beta)$ ($\beta > 1$). $N < \infty$ is the physical truncation, corresponding to a real system with finitely many excitations.

Theorem 2.5 (**[A]**)**.** *Key properties of the truncated partition function $Z_N(\beta)$:*

(a) *For all $\beta > 0$ and $N < \infty$, $Z_N(\beta)$ is an entire function of β (no poles);*

(b) *$Z_N(\beta) \to \zeta(\beta)$ pointwise as $N \to \infty$ (for $\beta > 1$);*

(c) *$Z_N(\beta) < \zeta(\beta)$ for all $\beta > 1$ and finite N;*

(d) *$\zeta(\beta)$ has a simple pole at $\beta = 1$, while $Z_N(\beta)$ is analytic at $\beta = 1$—this contrast is the key to the theory of quasi-black holes.*

2.3 KMS States and β [A]

Theorem 2.6 ([**A**], Bost-Connes 1995 [1])**.** *The KMS state structure of the BC algebra* $(\mathcal{A}_{\mathrm{BC}}, \sigma_t)$ *is completely determined by* β*:*

(1) ***High-temperature phase*** $\beta \le 1$*: the system has a unique KMS*$_\beta$ *state, with density matrix* $\rho \propto e^{-\beta H_{\mathrm{BC}}}$ *(unnormalized; the partition function diverges for* $\beta \le 1$*, understood as an appropriate limit);*

(2) ***Low-temperature phase*** $\beta > 1$*: there exist continuously infinitely many extremal KMS*$_\beta$ *states, parametrized by* $\hat{\mathbb{Z}}^* = \varprojlim(\mathbb{Z}/n\mathbb{Z})^*$ *(the abelianization of the absolute Galois group of* $\mathbb{Q}$*);*

(3) ***Phase transition point*** $\beta = 1$*: exists only in the* $N \to \infty$ *limit, corresponding to the pole of* $\zeta(\beta)$*;* ***the phase transition is smeared out for finite*** N *(*Z_N *has no pole).*

Corollary 2.7 ([**A**])**.** *The extremal KMS*$_\beta$ *states in the* $\beta > 1$ *phase are parametrized by elements of the Galois group* $\hat{\mathbb{Z}}^*$*, each corresponding to a distinct "phase." These phases carry information—they form the mathematical basis of the information-preservation mechanism.*

2.4 Crossover Width for Finite N [A]

Theorem 2.8 (Crossover Width, [**A**])**.** *For finite truncation* N*, define the crossover region as the range of parameters over which* $Z_N(\beta)$ *changes from "close to* $\zeta(\beta)$*" to "appreciably smaller than* $\zeta(\beta)$*." This can be computed precisely using the Euler-Maclaurin formula:*

For $\beta = 1 + \delta$ *(*$\delta > 0$ *small),*

$$Z_N(1+\delta) = \zeta(1+\delta) - \frac{N^{-\delta}}{\delta} + O\left(\frac{1}{N^{1+\delta}}\right).$$

The difference $Z_N(\beta) - \zeta(\beta)$ *is* $O(N^{-\delta}/\delta)$*, which is* $O(1)$ *when* $\delta \sim 1/\log N$*. The* ***crossover width*** *(exact) is therefore:*

$$\Delta\beta = \frac{\ln 2}{\ln N}.$$

Remark 2.9. The larger N, the narrower the crossover region, and the more the system "looks like" it has a sharp phase transition—but it never is sharp ($N < \infty$). This is a universal property of finite systems, independent of GR geometry.

3 Definition of Quasi-Black Holes

3.1 Definition [A*]

Definition 3.1 (Quasi-Black Hole, [**A***])**.** A **quasi-black hole** is a state of the BC system satisfying the following two purely BC conditions:

(1) Maximal truncation parameter: $N \gg 1$ (N is finite but much greater than 1);

(2) Effective inverse temperature in the crossover region:

$$\beta_{\mathrm{eff}} \in \left[1,\, 1+\Delta\beta\right] = \left[1,\, 1 + \frac{\ln 2}{\ln N}\right],$$

i.e., the partition function $Z_N(\beta_{\mathrm{eff}})$ is close to (but not equal to) $\zeta(\beta_{\mathrm{eff}})$:

$$Z_N(\beta_{\mathrm{eff}}) \approx \zeta(\beta_{\mathrm{eff}}), \quad \frac{Z_N(\beta_{\mathrm{eff}})}{\zeta(\beta_{\mathrm{eff}})} > 1 - \varepsilon(N),$$

where $\varepsilon(N) \to 0$ as $N \to \infty$.

3.2 Absence of a Sharp Phase Transition [A]

Proposition 3.2 (Absence of Sharp Horizon, [**A***])**.** *A quasi-black hole has no sharp event horizon. Argument:*

(1) *We identify a "sharp horizon" with the exact phase transition of the BC algebra at $\beta = 1$ (the ζ pole);*

(2) *The pole of $\zeta(\beta)$ only appears in the $N \to \infty$ limit ([**A**], Theorem 2.5(d));*

(3) *A quasi-black hole has $N < \infty$ (Definition 3.1, Condition 1);*

(4) *Therefore $Z_N(\beta)$ has no pole $\Rightarrow$ no exact phase transition $\Rightarrow$ no sharp horizon.*

Corollary 3.3 ([**A**])**.** *Finite N yields a **crossover region** of width $\Delta\beta = \ln 2/\ln N$, not a sharp horizon. The "horizon" is a product of the $N \to \infty$ limit (= GR black hole). For finite N there is only a smooth crossover region of width $\Delta\beta$.*

Remark 3.4 ($N \to \infty$ corresponds to GR black holes; finite N to quasi-black holes)*.*

$N \to \infty$ $\quad$ $Z_N \to \zeta$, pole exists, sharp phase transition, **GR black hole**

$N < \infty$ $\quad$ Z_N has no pole, crossover region of width $\Delta\beta$, **quasi-black hole**

Quasi-black holes physically exist in nature (N finite); GR black holes are a mathematical limit.

Remark 3.5 (Implications for Astronomical Observations)*.* The gravitational wave signals detected by LIGO/VIRGO [18] and the EHT shadow images [19, 20] are in excellent agreement with GR predictions—but this does not rule out quasi-black holes. For stellar-mass objects, $N \sim 10^{77}$, the crossover width is $\Delta\beta \sim 10^{-38}$, corresponding to a spatial scale far below the Planck length, and indistinguishable from GR predictions at current observational precision. The compact objects currently observed are therefore more likely to be quasi-black holes rather than exact GR black holes—the latter require $N = \infty$, a physically unrealizable limit. This distinction has no effect on any observable for stellar-mass objects, but it yields qualitatively different predictions in the terminal evaporation phase ($N \sim 1$) (see Secs. 6 and 10).

4 Formation of Quasi-Black Holes

4.1 Initial State [A*]

Proposition 4.1 (Initial State, [**A***])**.** *Before collapse, the BC description of the system (a star) is:*

- *A small number of states are excited: N_{init} is small (relative to the final state);*
- *$\beta_{\text{eff}} \gg 1$ (much greater than 1; the system is very cold);*
- *$Z_{N_{\text{init}}}(\beta) \ll \zeta(\beta)$ (far from the crossover region).*

This is a "cold" BC state: excited states are sparse, and the partition function is much smaller than $\zeta(\beta)$.

4.2 Excitation Driven by the Explicit Formula [A]

The mathematical core of the BC formation mechanism comes from the Riemann explicit formula:

Theorem 4.2 (Explicit Formula, [**A**], [29, 30])**.** *The Chebyshev function $\psi(x) = \sum_{p^k \le x} \log p$ satisfies the exact expansion:*

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2}\log\bigl(1 - x^{-2}\bigr),$$

where the sum runs over all non-trivial zeros ρ of $\zeta(s)$.

Equivalently, the von Mangoldt function satisfies (in the distributional sense):

$$\Lambda(n) + \sum_{\rho} n^{\rho-1} = 1,$$

where $\Lambda(n) = \log p$ if $n = p^k$ is a prime power, and $\Lambda(n) = 0$ otherwise.

Proposition 4.3 (Excitation Driven by the Matter Term, [**A***])**.** *The explicit formula is interpreted as the excitation dynamics of the BC system:*

- ***Matter term** $\Lambda(n) > 0$: nonzero only at $n = p^k$ (prime powers), corresponding to the excitation contribution of the "prime channels" in the BC system;*
- ***Zero correction term** $\sum_{\rho} n^{\rho-1}$: arising from the non-trivial zeros of ζ, this is the contribution of quantum fluctuations;*
- ***Constant term** 1: energy conservation constraint.*

*Physical interpretation ([**A***]): $\Lambda(n) > 0$ at prime powers is equivalent to "matter accumulating in the prime channels," which through the multiplicative structure of the BC algebra (i.e., Hecke operators) attracts more states to be excited, causing the effective number of degrees of freedom N to increase.*

4.3 The Process of Increasing N [A*]

Proposition 4.4 (N Increasing, [**A***])**.** *As matter accumulates (the BC description of classical gravitational collapse), the BC system evolves as follows:*

(1) *Initial state: N small, $\beta_{\text{eff}} \gg 1$, $Z_N(\beta) \ll \zeta(\beta)$;*

(2) *Prime-power channels with $\Lambda(n) > 0$ excite new states $|n\rangle$ into $\mathcal{H}_N$;*

(3) *N increases $\to Z_N(\beta)$ approaches $\zeta(\beta) \to$ the crossover region narrows ($\Delta\beta = \ln 2/\ln N$ decreases);*

(4) *The system looks increasingly "like" a sharp phase transition, but N remains finite $\to$ it never arrives.*

Remark 4.5. Step (4) is crucial: increasing N drives the system toward the $N = \infty$ limit, but it can never reach it. This is the mathematical origin of the "freezing" mechanism—no GR concept is needed.

4.4 Frozen State = Quasi-Black Hole [A*]

Proposition 4.6 (Freezing, [**A***])**.** *The process of increasing N ultimately reaches **freezing**:*

Once N has grown to some very large value $N_0 \gg 1$, the system satisfies the quasi-black hole definition (Sec. 3.1):

$$Z_{N_0}(\beta_{\text{eff}}) \approx \zeta(\beta_{\text{eff}}), \quad \beta_{\text{eff}} \in \left[1,\, 1 + \frac{\ln 2}{\ln N_0}\right].$$

*At this point Z_{N_0} has no pole (finite N_0), so the system **cannot** undergo an exact phase transition, the matter excitation stops growing—the system freezes at $N = N_0$.*

Frozen state = quasi-black hole.

Remark 4.7 (Summary of Formation Mechanism). Initial state (N small, cold) $\xrightarrow{\Lambda(n)>0 \text{ prime excitation}}$ N increases $\xrightarrow{Z_N\to\zeta}$ approaches the crossover region $\xrightarrow{N<\infty,\, Z_N \text{ has no pole}}$ freezes $\to$ quasi-black hole ($N = N_0 \gg 1$)

Throughout: no Schwarzschild radius, no Tolman temperature, no horizon crossing.

5 Properties of Quasi-Black Holes

5.1 Absence of Singularities [A]

Theorem 5.1 (Absence of Singularities, [**A**]). *The truncation $n \geq 1$ of the BC system (Hilbert space basis starting from $|1\rangle$) guarantees:*

The minimum eigenvalue of the BC Hamiltonian is $\log 1 = 0$ *(the ground state), which is finite. The BC analogue of GR curvature, $\kappa(2)$, is:*

$$\kappa(2) = \log\frac{Z(2)}{Z(4)} = \log\frac{\zeta(2)}{\zeta(4)} = \log\frac{\pi^2/6}{\pi^4/90} = \log\frac{15}{\pi^2} = \log\frac{9}{8} + O(1),$$

*which is finite—**no singularity**.*

Remark 5.2. The absence of singularities is a mathematical consequence of $n \geq 1$ (integers start at 1), not an externally imposed regularization. This mechanism is completely different from LQG (area quantization resolves singularities) or string theory (string regularization).

5.2 Discrete Entropy $S_{\mathrm{BC}} = \psi(N)$ [A*]

5.2.1 Truncated Galois Group

Definition 5.3 (Truncated Galois Group, [**A***]). The extremal KMS_β states in the BC low-temperature phase are parametrized by $\hat{\mathbb{Z}}^* = \varprojlim(\mathbb{Z}/n\mathbb{Z})^*$ ([**A**]). After truncating to finite N, the effective symmetry group is:

$$\hat{\mathbb{Z}}^*_N = (\mathbb{Z}/\mathrm{lcm}(1,2,\ldots,N)\mathbb{Z})^*,$$

with order

$$|\hat{\mathbb{Z}}^*_N| = \varphi(\mathrm{lcm}(1,2,\ldots,N)).$$

5.2.2 Exact Discrete Entropy

Definition 5.4 (BC Entropy, [**A***]). The **Bost-Connes entropy** is defined as the total complexity of the truncated system:

$$S_{\mathrm{BC}}(N) := \log\mathrm{lcm}(1,2,\ldots,N) = \psi(N).$$

Here ψ is the Chebyshev function. Note that $\log\mathrm{lcm}(1,\ldots,N) = \psi(N)$ is an exact identity (see below).

Remark 5.5 (Why lcm rather than φ(lcm)). The order of the truncated Galois group $\hat{\mathbb{Z}}^*_N$ is $\varphi(\mathrm{lcm}(1,\ldots,N))$, but $\log\varphi(m) = \log m + \sum_{p|m}\log(1-1/p) \neq \log m$. Specifically, $\log\varphi(\mathrm{lcm}) \approx \psi(N) - \log\log N - M$ (where M is the Meissel-Mertens constant), a discrepancy of about 5.2 for $N = 10^{77}$. The correct definition of BC entropy uses lcm itself (total period complexity), not the order of its unit group.

Theorem 5.6 ([**A**], Number Theory). $\psi(N) = \log\mathrm{lcm}(1,\ldots,N) = \sum_{p^k\leq N}\log p$ *is the Chebyshev ψ function, satisfying:*

(a) ***Prime number theorem (PNT)****:* $\psi(N) = N + O(N^{1/2+\varepsilon})$ *(asymptotically);*

(b) ***Exact expansion*** *(Riemann explicit formula [31]):*

$$\psi(N) = N - \sum_\rho \frac{N^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log\bigl(1 - N^{-2}\bigr)\,,$$

where the sum runs over all non-trivial zeros ρ *of* $\zeta(s)$*;*

(c) ***Strict monotonicity****:* $\psi(N+1) - \psi(N) = \Lambda(N+1) \geq 0$*, with strict inequality* > 0 *when* $N+1$ *is a prime power;*

(d) $\psi(1) = 0$ *(ground state has no entropy);* $\psi(N) \sim N$ *as* $N \to \infty$*.*

Remark 5.7 (BC Entropy vs. GR Entropy (Continuum Approximation)). **Exact BC entropy** (no GR input):

$$S_{\mathrm{BC}}(N) = \psi(N) = N - \sum_\rho \frac{N^\rho}{\rho} - \log(2\pi) - \cdots$$

Bekenstein-Hawking entropy [7] (continuum approximation, $N \to \infty$): $S_{\mathrm{BH}} = 4\pi N$.

Relation: $S_{\mathrm{BH}} \approx 4\pi\,\psi(N)$ (under the continuum approximation). The factor 4π is a geometric contribution from the continuum limit, not an intrinsic number of the BC algebra. The exact entropy $\psi(N)$ contains ζ-zero corrections—**unique to BC**, inaccessible to any continuum theory.

5.3 Four Laws of Thermodynamics (Analogy to [8]) [A]

Zeroth Law (Thermal Equilibrium) [A] The KMS condition (by definition) guarantees that all prime channels in the system share a unique global β, i.e., the system is in global thermodynamic equilibrium. This is a **rigorous algebraic result** (the KMS condition is itself the definition of equilibrium), independent of the concept of heat flow in continuum geometry.

First Law (Energy Conservation) [A*] The internal energy of the BC system (by definition):

$$E(\beta, N) = \langle H_{\mathrm{BC}} \rangle_{\beta,N} = \frac{\sum_{n=1}^{N} (\log n) \cdot n^{-\beta}}{Z_N(\beta)}.$$

First law (thermodynamic identity, [**A***]):

$$dE = T_{\mathrm{BC}}\, dS_{\mathrm{BC}} = T_{\mathrm{BC}}\, d\psi(N).$$

Second Law (Non-Decrease of Entropy) [A] By the strict monotonicity of $\psi(N)$ (Theorem 5.6(c)):

$$\frac{dS_{\mathrm{BC}}}{dN} = \frac{d\psi}{dN} = \Lambda(N) \geq 0.$$

Entropy is non-decreasing as N increases—**this is a theorem of number theory, requiring no assumption from statistical physics**.

Third Law (Unattainability) [A] Both extremes are unattainable:

- $\beta_{\mathrm{eff}} = 1$ (the phase transition point): requires $N = \infty$, unreachable for a system with finite N;
- $\beta_{\mathrm{eff}} = \infty$ (absolute zero): requires $N = 1$ and $\beta \to \infty$, approached asymptotically but never reached.

5.4 BC Temperature [A*]

Definition 5.8 (BC Temperature, [**A***])**.** The BC temperature is defined from the thermodynamic first law $dE = T_{\mathrm{BC}}\, dS_{\mathrm{BC}}$:

$$T_{\mathrm{BC}} = \frac{dE}{d\psi(N)}.$$

This is a purely BC definition, independent of the Hawking derivation or GR geometry.

Proposition 5.9 (Temperature Calculation, [**A***])**.** *Using $d\psi \approx dN$ (the leading PNT term, [**A**]) and the explicit expression for $E(\beta, N)$:*

$$T_{\mathrm{BC}}(N) = \left(\frac{dE}{dN}\right) \Big/ \left(\frac{d\psi}{dN}\right) \approx \frac{dE/dN}{\Lambda(N)}.$$

*For large N with β_{eff} in the crossover region, T_{BC} is determined self-consistently by $Z_N(\beta)$ (see Sec. 9 for the correspondence between BC and GR)—**without invoking the concept of Hawking radiation**.*

Remark 5.10 (Purely BC Nature of the Temperature)*.* $T_{\mathrm{BC}} = dE/d\psi$ is a purely BC definition; $E = \langle H_{\mathrm{BC}} \rangle$ is a quantum-mechanical quantity on Hilbert space, independent of any continuum GR concept. The correspondence between BC temperature and Hawking temperature is given in Sec. 9.

5.5 Information Preservation [A*]

Proposition 5.11 (Information is not Lost, [**A***])**.** *Quasi-black holes do not give rise to an information paradox:*

(1) *Finite $N \Rightarrow \beta_{\mathrm{eff}} > 1$ (in the low-temperature phase, never crossing $\beta = 1$, Proposition 3.2);*

(2) *$\beta_{\mathrm{eff}} > 1 \Rightarrow$ continuously many extremal KMS states exist (Theorem 2.3-(2));*

(3) *The extremal states are parametrized by the Galois labels of $\hat{\mathbb{Z}}_N^*$ (Definition 5.3);*

(4) *Galois labels carry information $\Rightarrow$ information is encoded in the algebraic structure ([**A***]);*

(5) *No sharp horizon (Proposition 3.2) $\Rightarrow$ no classical one-way boundary trapping information;*

(6) *The evaporation process $N : N_0 \to 1$, at each step the Galois group shrinks, and information is **gradually released** into the radiation ([**A***]).*

6 Evaporation of Quasi-Black Holes

6.1 Mechanism: KMS Detailed Balance [A*]

Theorem 6.1 (KMS Detailed Balance, [**A**]; [5])**.** *A KMS$_\beta$ state satisfies detailed balance [4, 1]: for any two states $|m\rangle$, $|n\rangle$ (with $E_m > E_n$), the ratio of the de-excitation rate $\Gamma(m \to n)$ to the excitation rate $\Gamma(n \to m)$ is*

$$\frac{\Gamma(m \to n)}{\Gamma(n \to m)} = e^{\beta(E_m - E_n)}.$$

This is a direct consequence of the KMS condition, not an additional assumption.

Proposition 6.2 (Direction of Evaporation, [**A***])**.** *The net direction of radiation from a quasi-black hole is determined by KMS detailed balance. For the transition from the highest excited state $|N\rangle$ to $|N-1\rangle$:*

(1) *Energy difference* $\Delta E = \log N - \log(N-1) = \log(N/(N-1)) > 0$ *(**[A]**);*

(2) *Detailed balance gives the de-excitation/excitation ratio* $\Gamma_\downarrow/\Gamma_\uparrow = (N/(N-1))^\beta > 1$ *(Theorem 6.1);*

(3) $\beta_{\text{eff}} > 1$ *(quasi-black holes are in the low-temperature phase, Proposition 3.2);*

(4) *Therefore **de-excitation dominates**: the net effect is* $N \to N-1$*, releasing energy* ΔE*.*

Radiation requires no horizon, no vacuum fluctuations, and no Hawking semi-classical derivation—it is the natural de-excitation of a KMS state at $\beta > 1$*.*

Remark 6.3. Each de-excitation $N \to N-1$ releases one Galois label (the minimal unit of information), reducing the system complexity $\mathrm{lcm}(1,\ldots,N) \to \mathrm{lcm}(1,\ldots,N-1)$. When N is a prime power, $\psi(N) - \psi(N-1) = \log p > 0$ (a step-down in entropy).

6.2 Evaporation Rate [A*]

Proposition 6.4 (Evaporation Rate, [**A***])**.** *The evaporation rate of a quasi-black hole is given by the Gibbs occupation probability (**[A]**: Gibbs form of the KMS state):*

$$\frac{dN}{dt} = -P(N), \qquad P(N) = \frac{N^{-\beta_{\text{eff}}}}{Z_N(\beta_{\text{eff}})}.$$

***Derivation**:*

(a) $P(N)$ *is the occupation probability of state* $|N\rangle$ *in the* KMS_β *distribution (**[A]**, Gibbs distribution);*

(b) *Detailed balance ensures that the de-excitation rate is proportional to the occupation probability (Theorem 6.1);*

(c) *The proportionality constant is taken to be 1 (in units of BC time* σ_t*, **[A*]**).*

***Qualitative behavior**: For large* N*,* $P(N) \sim N^{-\beta}/\zeta(\beta) \ll 1$ *(evaporation extremely slow); as* N *decreases,* $P(N)$ *increases (evaporation accelerates); at* $N = 1$*,* $P(1) = 1/Z_1 = 1$*, but there is no lower state to de-excite to—radiation ceases.*

6.3 Page Curve [A*]

Proposition 6.5 (Page Curve, [**A***])**.** *The BC Page curve of a quasi-black hole is:*

$$S_{\text{BC}}(t) = \psi(N(t)),$$

where $N(t)$ *satisfies* $dN/dt = -P(N)$ *(Sec. 6.2).*

By the strict monotonicity of ψ *(**[A]**) and the monotonic decrease of* $N(t)$ *(a consequence of **[B]**):* $S_{\text{BC}}(t)$ *is strictly monotonically decreasing.*

*Radiation entropy (total entropy conservation, **[A*]**):* $S_{\text{rad}}(t) \approx \psi(N_0) - \psi(N(t))$*, monotonically increasing.*

Remark 6.6 (Uniqueness of the BC Page Curve). The BC Page curve follows directly from the monotonicity of $\psi(N)$, **without requiring**: the island formula, Euclidean gravitational path integrals, or holographic duality. Monotonicity is a theorem of number theory ([**A**]), not an input from quantum gravity.

7 Endpoint of Quasi-Black Holes

7.1 Terminal Phase: $N \to 1$, Radiation Fades [A*]

Proposition 7.1 (Terminal Phase, [**A***])**.** *As $N(t)$ decreases to $N \sim 1$:*

- $Z_1(\beta) = 1$ *(only the ground state $|1\rangle$, partition function identically 1);*
- *Evaporation rate $P(1) = 1/Z_1 = 1$, but there is no lower state to de-excite to $\to$ radiation **fades**;*
- $\beta_{\text{eff}} \to \infty$ *(third law: unattainable, approached asymptotically);*
- $S_{\text{BC}} = \psi(1) = 0$ *(ground state has no entropy).*

Remark 7.2. "Fading" is a direct consequence of the BC third law: $\beta = \infty$ is unattainable, so radiation does not stop abruptly at some moment but decays exponentially to zero. This is qualitatively different from the GR/Hawking "terminal burst."

7.2 Planck Remnant: the $N = 1$ Ground State [A*]

Proposition 7.3 (Planck Remnant, [**A***])**.** *The end state of evaporation is the **Planck remnant**:*

$$N = 1, \quad Z_1(\beta) = 1, \quad S_{\text{BC}} = \psi(1) = 0.$$

Properties:

(1) $S = 0$*: no thermodynamic degrees of freedom, no radiation;*

(2) ***Stable****: $N = 1$ is the ground state of the BC system, with no lower state and no evaporation channel;*

(3) *The third law guarantees that $\beta = \infty$ (zero temperature) cannot be reached, so the remnant is not an absolute-zero ground state but a "quantum remnant" at $N = 1$: it does not radiate but is not at zero temperature.*

Remark 7.4. The Planck remnant ($N = 1$) is the natural end state of the BC framework, not an externally imposed condition. Its mass in the continuum approximation is $\sim M_{\text{Pl}}$ (see Sec. 9 for the correspondence), and it undergoes no further evaporation.

7.3 Fate of Information: Fully Released During Evaporation [A*]

Proposition 7.5 (Fate of Information, [**A***])**.** *The complete trajectory of information:*

(1) *Quasi-black hole forms: information is encoded in the Galois labels of $\hat{\mathbb{Z}}^*_{N_0}$ (N_0 degrees of freedom);*

(2) *Each step $N \to N - 1$: one Galois label is released, and information **gradually** enters the radiation;*

(3) *Evaporation complete ($N : N_0 \to 1$): all $N_0 - 1$ degrees of freedom have been released;*

(4) *Remnant ($N = 1$): $S = 0$, no Galois structure, **carries no information**.*

Information is not in the remnant—it is entirely in the evaporation radiation.

Remark 7.6 (Summary of the Life Cycle of a Quasi-Black Hole). Initial state (N small, cold) $\xrightarrow{\text{formation}}$ quasi-black hole ($N = N_0$, frozen) $\xrightarrow{\text{evaporation}}$ decreasing ($N_0 \to 1$) $\xrightarrow{\text{endpoint}}$ Planck remnant ($N = 1$, $S = 0$)

$$\begin{aligned}
\text{Core quantities throughout:} \quad & Z_N(\beta) = \sum_{n=1}^{N} n^{-\beta} \\
\text{Entropy:} \quad & S_{\mathrm{BC}} = \psi(N) \\
\text{Temperature:} \quad & T_{\mathrm{BC}} = dE/d\psi \\
\text{Not needed:} \quad & \text{Tolman, Schwarzschild, Hawking (these appear as approximations} \\
& \text{in Sec. 9)}
\end{aligned}$$

8 Generalization: The Universe as a Quasi-Black Hole

The definition of a quasi-black hole (Definition 3.1) imposes no upper bound on N. Extending the framework to cosmological scales yields a unified picture.

8.1 Complete Evolution Equation for $N(t)$ [A*]

Proposition 8.1 ($N(t)$ Dynamics, [**A***]). *The truncation parameter $N(t)$ of a quasi-black hole (or any BC subsystem) satisfies*

$$\frac{dN}{dt} = R_{\mathrm{in}}(t) - P(N),$$

where $R_{\mathrm{in}}(t)$ is the accretion rate driven by external matter, and $P(N) = N^{-\beta}/Z_N(\beta)$ is the de-excitation probability of the highest state $|N\rangle$ under detailed balance (Proposition 6.2). $N(t)$ first increases and then decreases, with a peak $N_{\max}$:

(i) ***Accretion phase*** *($R_{\mathrm{in}} > P$): $dN/dt > 0$, N increases;*

(ii) ***Peak*** *($R_{\mathrm{in}} = P$): $N = N_{\max}$, closest to GR ($\Delta\beta$ smallest);*

(iii) ***Evaporation phase*** *($R_{\mathrm{in}} < P$): $dN/dt < 0$, N decreases to 1.*

Remark 8.2 (BC Interpretation of the GR Area Theorem). Within the BC framework, the GR area theorem is reinterpreted as corresponding to the BC accretion phase $R_{\mathrm{in}} > P$; it is an approximate behavior during one phase, not a fundamental law. In BC, $\psi(N)$ can increase or decrease—evaporation requires no "violation" of any theorem.

8.2 The Universe as the Largest Quasi-Black Hole [A*]

Proposition 8.3 (The Universe as a Quasi-Black Hole, [**A***]). *The observable universe itself is a quasi-black hole:*

- $N_{\mathrm{univ}} \approx 10^{122}$ *(total degrees of freedom of the universe; estimated from the Bekenstein bound [7] $S \leq 2\pi ER$ applied to the Hubble volume of the observable universe in the continuum approximation);*
- *At the Big Bang, β passed through the crossover region ($\beta \approx 1$); thereafter $\beta(t) = T_{\mathrm{Pl}}/T_{\mathrm{CMB}}(t)$ increases monotonically;*
- *Currently $\beta_{\mathrm{now}} = T_{\mathrm{Pl}}/T_{\mathrm{CMB}} \approx 10^{32}$ (extremely cold, deep in the classical phase; this value uses the CMB temperature $T_{\mathrm{CMB}} \approx 2.7\,\mathrm{K}$ in the continuum approximation).*

Stellar-mass quasi-black holes are subsystems of the universe, the largest quasi-black hole: $N_{\rm BH} \le N_{\rm univ}$.

Corollary 8.4 ($N_{\rm univ}$ Conservation and Subsystem Evaporation, **[A*]**)**.** *The cosmic quasi-black hole differs from stellar-mass quasi-black holes in a fundamental respect:*

- *A stellar-mass quasi-black hole is a subsystem of the universe; an external thermal reservoir exists. Evaporation means* $N_{\rm BH}$ *decreases while the environment absorbs the released degrees of freedom.*
- *The cosmic quasi-black hole has no external system;* $N_{\rm univ}$ *is fixed after the Big Bang and admits no "evaporation" channel.*
- *Subsystem truncation parameters may increase or decrease (accretion or evaporation), subject to the conservation constraint* $\sum_i N_i \le N_{\rm univ}$.

The thermodynamic evolution of the universe is therefore not evaporation but internal redistribution of N*: early epoch* ($\beta \approx 1$*, roughly uniform occupation)* $\to$ *structure formation (*N *concentrated in subsystems)* $\to$ *structure evaporation (*N *redispersed)* $\to$ *late epoch (*$\beta \gg 1$*, dilute uniform distribution). The Third Law guarantees that this process has no endpoint.*

8.3 Inaccessibility of ζ [A*]

Proposition 8.5 (Inaccessibility of ζ, **[A*]**)**.** $N_{\rm univ}$ *finite* $\Rightarrow Z_N(\beta) \ne \zeta(\beta)$ *for all* β. $\zeta(\beta)$ *is the* $N \to \infty$ *limit, not a physical quantity:*

- $\beta \approx 1$ *(Big Bang):* $Z_N(1) = H_N \approx \ln N \approx 281$*, while* $\zeta(1) = \infty$*;*
- $\beta \approx 10^{32}$ *(present):* $Z_N \approx \zeta$*, difference* $\sim 10^{-10^{32}}$ *(exponentially small);*
- $\beta \to \infty$ *(far future):* $Z_N \to 1$, $\zeta \to 1$*, difference* $\to 0$*, but never zero.*

Corollary 8.6 (No Heat Death, **[A*]**)**.** *A cosmological consequence of the BC third law (*$\beta = \infty$ *unattainable): the universe can never reach absolute zero temperature (*$\beta = \infty$*), so there will always be small thermal fluctuations—there is no heat death in the classical sense.*

Proposition 8.7 (GR's Four Fundamental Problems and Their BC Resolution, **[A*]**)**.** *GR's four fundamental difficulties all originate from the* $N \to \infty$ *limit:*

GR Difficulty	***Product of*** $N \to \infty$	***Finite*** N ***Resolution***
Singularity	*Pole of* $\zeta(\beta)$	Z_N *has no pole*
Information paradox	*Sharp horizon*	*Crossover region of width* $\Delta\beta > 0$
Cosmological constant problem (10^{122}*)*	*Infinite degrees of freedom*	N *finite and large*
Heat death	$\beta = \infty$ *reachable*	*Third law: unreachable*

Z_N *(finite) is the physics;* ζ *(the limit) is not.*

9 Comparison with Continuum Theories

Remark 9.1 (**Nature of This Section**). This section (Sec. 9) is the **only** place in this paper where continuum GR concepts such as Tolman temperature, Schwarzschild radius, and Hawking radiation appear. They appear here as "continuum approximations to BC in the $N \to \infty$ limit," to demonstrate how BC recovers GR results.

9.1 Continuum Limit: BC → GR ($N \to \infty$)

N–Mass Correspondence (Continuum Approximation) [A*] In the continuum approximation, the relationship between the effective number of degrees of freedom N and the gravitational mass M comes from the Bekenstein-Hawking area formula (a GR result):

$$N \approx \frac{M^2}{M_{\mathrm{Pl}}^2}, \quad M_{\mathrm{Pl}} = \sqrt{\frac{\hbar c}{G}} \approx 2.2 \times 10^{-8}\,\mathrm{kg}.$$

Continuum Limit of Entropy [A*] For $N \to \infty$, $\psi(N) \sim N$ (PNT); introducing the 4π solid-angle factor upon taking the continuum limit:

$$S_{\mathrm{BH}} = \frac{A}{4G} = \frac{4\pi r_s^2}{4\ell_{\mathrm{Pl}}^2} = 4\pi \left(\frac{M}{M_{\mathrm{Pl}}}\right)^2 = 4\pi N \approx 4\pi\, \psi(N).$$

The Bekenstein-Hawking entropy is therefore the continuum approximation to the exact BC entropy:

$$S_{\mathrm{BH}} = 4\pi\, \psi(N) \approx 4\pi N, \quad (N \to \infty, \text{ continuum approximation})$$

The factor 4π is a continuum geometric factor, not an intrinsic number of the BC algebra.

Continuum Limit of Temperature [A*] From $N = M^2/M_{\mathrm{Pl}}^2$ and $dE = T_{\mathrm{BC}}\, d\psi \approx T_{\mathrm{BC}}\, dN$:

$$T_{\mathrm{BC}} = \frac{dE}{dN} = \frac{dM}{d(M^2/M_{\mathrm{Pl}}^2)} = \frac{M_{\mathrm{Pl}}^2}{2M}.$$

The Hawking temperature (continuum GR):

$$T_{\mathrm{Hawking}} = \frac{\hbar c^3}{8\pi G M} = \frac{M_{\mathrm{Pl}}^2}{8\pi M}, \quad \text{(Schwarzschild black hole, [}\mathbf{A}\text{]for GR)}$$

therefore

$$T_{\mathrm{BC}} = 4\pi \times T_{\mathrm{Hawking}}.$$

The factor 4π **dually compensates** between temperature and entropy: $T_{\mathrm{BC}} \cdot S_{\mathrm{BC}} = T_{\mathrm{Hawking}} \cdot S_{\mathrm{BH}}$.

Continuum Limit: Explicit Formula and the Einstein Equations [B] (Physically motivated; awaiting rigorous justification.) The BC explicit formula $\Lambda(n) + \sum_\rho n^{\rho-1} = 1$, in the $N \to \infty$ continuum limit, yields an energy-curvature relation—corresponding to the Einstein equations $G_{\mu\nu} = 8\pi T_{\mu\nu}$ ([**B**]).

Tolman Temperature as a Continuum Approximation [A*] The GR black hole criterion "$T_{\mathrm{local}} \to T_{\mathrm{Pl}}$" is equivalent, in the continuum approximation, to the BC criterion "$Z_N(\beta) \approx \zeta(\beta)$":

$$T_{\mathrm{local}}(r) = \frac{T_\infty}{\sqrt{1 - r_s/r}} \to T_{\mathrm{Pl}} \quad \Leftrightarrow \quad \beta_{\mathrm{eff}} \to 1^+ \quad \Leftrightarrow \quad Z_N(\beta) \to \zeta(\beta).$$

The Tolman formula [11] is the continuum GR version of the BC crossover condition.

9.2 Comparison Table: BC vs. Continuum Theories

Feature	**BC (this work, exact)**	**GR (continuum approx.)**	**LQG** [24]	**String/fuzzball** [25]
Fundamental parameter	N (truncation, BC-intrinsic)	M, r_s (mass/radius)	area quanta	string/brane d.o.f.
Horizon	None (Z_N no pole, [**A**])	Sharp (ζ pole, $N \to \infty$)	quantum correction	fuzz substitute
Singularity	None ($n \geq 1$, [**A**])	Exists	resolved	resolved
Entropy	$\psi(N)$ (exact, [**A***])	$4\pi N$ (continuum approx.)	$A/(4\ell_{\mathrm{Pl}}^2)$	microstate count [26]
Entropy correction	ζ zeros (**unique**)	None	$\log A$	log microstates
Temperature	$dE/d\psi$ (BC definition)	Hawking [9, 10]	–	–
Information	Gradually released (Galois labels)	Paradox	Unitary (not rigorous)	Unitary (holographic [14])
Final state	$N = 1$, $S = 0$ (fade)	$M = 0$ (burst)	Remnant (some schemes)	Remnant
New physics	**Not required** ($\mathbb{Z}$ only)	–	quantum geometry	strings/D-branes/holography
Observable window	PBH terminal radiation	PBH burst	sub-Planck scale	10^{26} dimensions

Three unique contributions of the BC framework:

(1) **Exact discrete entropy** $\psi(N)$: ζ-zero corrections; no other scheme can provide these;

(2) **Absence of singularities guaranteed by** $n \geq 1$: not an additional assumption, but an algebraic fact about the integers;

(3) **Only** $\mathbb{Z}$ **required**: no new matter, no new symmetry, no extra dimensions—the algebraic structure of the integers suffices to describe black hole thermodynamics. By contrast, gravastars [21] require a dark-energy core, fuzzballs require string-theoretic extra dimensions, and analogue gravity schemes [22] require an acoustic analogue.

10 Observational Predictions

Remark 10.1 (**Location of Honest Assessment**). All quantitative estimates below rely on the correspondence $N = M^2/M_{\mathrm{Pl}}^2$ from Sec. 9 (continuum approximation, [**A***]).

10.1 Stellar Scale: Indistinguishable from GR [A*]

(The following uses the continuum approximation of Sec. 9.)

For stellar-mass quasi-black holes ($M \sim 10 M_\odot$, $N \sim 10^{76}$):

- Crossover width (BC parameter): $\Delta\beta = \ln 2/\ln N \sim 10^{-76}$;
- Corresponding spatial width of the crossover region in continuum geometry: $\Delta r \sim \ell_{\rm Pl}/N \sim 10^{-78}\,$m;
- This is smaller than the proton radius ($\sim 10^{-15}\,$m) by a factor of 10^{-63}—completely undetectable;
- EHT shadow sizes and gravitational wave signals differ from those of a GR black hole by $O(N^{-1}) \approx 0$.

Conclusion: Stellar-mass quasi-black holes are indistinguishable from GR black holes in all current and foreseeable observations.

10.2 PBH Terminal Phase: Fade vs. Burst (the Unique Observable Window) [A*]

Proposition 10.2 (PBH Terminal Phase, [**A***])**.** *Primordial black holes (PBHs) in their terminal phase ($N \to 1$) represent the largest discrepancy between BC and GR:*

	GR/Hawking	*BC (this work)*
Terminal temperature	$T_H \to \infty$ *(diverges)*	$T_{\rm BC} \to$ *finite* ($dE/d\psi$ *bounded)*
Terminal radiation	*burst* (γ*-ray flash)*	*fade (exponential decay)*
Final state	$M = 0$*, complete evaporation*	$N = 1$*, Planck remnant*
Observable signal	*burst: search for brief* γ *flashes*	*fade: search for "disappearing" PBHs*

Detectors: Fermi-LAT (γ-rays), HAWC (very high energy), future CTA.

10.3 Honest Assessment

Prediction	BC vs. GR Difference	Detectability	Prerequisites
Stellar-scale shadow	$\sim 10^{-78}$ m	**Not detectable**	–
Gravitational wave echoes [23]	$\sim e^{-S_{\rm BH}} \approx 0$	**Not detectable**	–
PBH terminal: fade vs. burst	Qualitative difference	**Observable in principle**	PBH must exist
PBH mass range	$\sim 4\pi$ offset	**Observable in principle**	PBH must exist
$\psi(N)$ exact Page curve	zero corrections	**Currently not detectable**	PBH must exist

Core honesty: if PBHs do not exist, none of the predictions in this paper can be tested. The existence of PBHs itself has no direct evidence (only indirect constraints). This is a fundamental limitation of the BC black hole theory, and we do not evade it.

11 Conclusions

11.1 Life Cycle Review

1. **Formation (N increases)**: The initial state has few excitations: N small, $\beta_{\text{eff}} \gg 1$ (cold); the matter term $\Lambda(n) > 0$ at prime powers drives more states to be excited, increasing N; increasing $N \to Z_N \to \zeta \to$ the crossover region narrows $\to$ approaches the phase transition but never reaches it; freezes at $N_0 \gg 1$, forming a quasi-black hole.

2. **Existence (finite N)**: N_0 is very large but finite; Z_{N_0} has no pole, no sharp horizon, no singularity; $S_{\text{BC}} = \psi(N_0)$ (carries Galois information); $T_{\text{BC}} = dE/d\psi$ (extremely low; system is very cold).

3. **Evaporation (N decreases)**: The highest excited state $|N\rangle$ de-excites; $N : N_0 \to 1$ (decreasing step by step); evaporation rate $dN/dt = -N^{-\beta}/Z_N(\beta)$ (small, fading); each step releases one Galois label (information gradually enters the radiation); $S_{\text{BC}}(t) = \psi(N(t))$ strictly monotonically decreasing (Page curve).

4. **Endpoint (Planck remnant)**: $N = 1$, $Z_1 = 1$, $S_{\text{BC}} = 0$, radiation fades; stable ground state, no evaporation, carries no information; all information has been released during evaporation.

11.2 Core Formulae

Remark 11.1 (Core of BC Black Hole Physics (no GR input)). **Truncated partition function**:

$$Z_N(\beta) = \sum_{n=1}^{N} n^{-\beta}, \qquad \lim_{N\to\infty} Z_N(\beta) = \zeta(\beta)$$

Exact discrete entropy:

$$S_{\text{BC}}(N) = \psi(N) = N - \sum_{\rho} \frac{N^\rho}{\rho} - \log(2\pi) - \cdots$$

BC temperature (defined from the first law):

$$T_{\text{BC}} = \frac{dE}{d\psi}, \qquad E = \langle H_{\text{BC}} \rangle = \frac{\sum_{n=1}^{N} (\log n)\, n^{-\beta}}{Z_N(\beta)}$$

Crossover width (exact, from Euler-Maclaurin):

$$\Delta\beta = \frac{\ln 2}{\ln N}$$

GR is the continuum limit ($N \to \infty$): $\psi(N) \to N \xrightarrow{\times 4\pi} S_{\text{BH}}$; $\quad T_{\text{BC}} \xrightarrow{\div 4\pi} T_{\text{Hawking}}$; $\quad 4\pi$ is a continuum geometric factor

11.3 Open Problems

The purely BC derivations in this paper have resolved most of the open problems present in earlier versions: N is the BC truncation parameter (no M–N mapping needed), evaporation is driven by KMS detailed balance ([**A***], Theorem 6.1), the quasi-black hole definition uses $Z_N \approx \zeta$ in place of the Tolman temperature (no localization of β_{eff} required), and 4π is a continuum geometric factor (need not be derived from BC). The remaining core problems are:

1. **Radiation spectrum**: KMS detailed balance determines the direction and rate of evaporation ([**A***]), but the energy spectrum (frequency distribution) of the radiation has not yet been derived from the BC algebra. A precise quantitative comparison with the Hawking radiation spectrum is the next task.
2. **Accretion rate** $R_{\text{in}}(t)$: In the $N(t)$ evolution equation (Proposition 8.1), the accretion term R_{in} depends on the external environment; the BC framework has not yet provided a way to compute R_{in} from first principles.

A Limitations

A.1 Correspondence between N and Gravitational Mass M

N is the BC truncation parameter and the fundamental quantity of this framework (Sec. 2.2). In the continuum approximation, $N \approx M^2/M_{\text{Pl}}^2$ (Sec. 9), but this correspondence comes from GR, not from a BC derivation. Numerical estimates (crossover width, PBH mass range, etc.) inherit the uncertainty of this correspondence.

A.2 Proportionality Constant in the Evaporation Rate

The direction of evaporation (de-excitation dominates) is rigorously determined by KMS detailed balance ([**A**], Theorem 6.1). In the evaporation rate formula $dN/dt = -P(N)$, $P(N)$ is determined by the Gibbs distribution ([**A**]), but taking the proportionality constant to be 1 (in units of BC time) is a convention ([**A***]); a rigorous derivation of the radiation spectrum from the BC algebra and a precise quantitative comparison with Hawking radiation are still lacking.

A.3 Predictions Contingent on PBHs

All observable predictions depend on the existence of primordial black holes (PBHs) [27]—for which there is currently no direct evidence. If PBHs do not exist, this paper has no testable predictions.

A.4 Scope of the BC Framework

The BC algebra is a purely mathematical object. Using BC to describe gravitational systems requires the assumptions that (i) the quantum-gravity degrees of freedom of nature are described by $\ell^2(\mathbb{Z}_{>0})$, and (ii) the gravitational Hamiltonian corresponds to H_{BC}. These assumptions have no independent verification to date.

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